

V-VIII Semester NEP Syllabus for Core courses

1. Computer Networks

B. OVERVIEW OF THE COURSE:

Name of Course	Computer Networks
Offering Department	COMPUTER SCIENCE & ENGINEERING

OVERVIEW:

The Computer Networks course aims to provide students with a comprehensive understanding of network architecture, protocols, and communication models. It covers fundamental concepts such as OSI and TCP/IP models, data transmission, network topologies, and routing algorithms. Students will learn about wired and wireless networks, LAN, WAN, and internet technologies. The course explores network security, congestion control, and error detection techniques to ensure reliable data communication. Hands-on labs will involve network simulation tools, socket programming, and configuring network devices. Students will also study protocols like HTTP, FTP, TCP, UDP, and IPv6, gaining insights into real-world networking applications. Advanced topics include cloud computing, IoT networking, and cybersecurity fundamentals. By the end of the course, students will be proficient in network design, troubleshooting, and performance optimization.

C. SYLLABUS

COURSE NO	TITLE OF THE COURSE	COURSE STRUCTURE	PRE-REQUISITE
CO CSC 501	Computer Networks	3L - 0T - 2P	None
COURSE OUTCOMES (COs) After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills, and attitudes: 1. Acquire a thorough understanding of state-of-the-art in modern network architecture. 2. Able to distinguish different techniques of error detection and correction and medium access control. 3. Learn addressing and forwarding mechanisms and solving relevant problems. 4. Learn and analyze various transport layer protocols like UDP, TCP, and various mechanisms to control TCP congestion. 5. Understand the purpose of application layer and the concepts of network security			
COURSE CONTENTS Unit I Introduction: Computer Networks, Data Flow, Network Topology, Reference Models – OSI, TCP/IP Unit II Physical Layer: Analog and Digital Data, Data Rate Limits, Performance Analysis, Line Coding, Bandwidth Utilizations, Transmission Media, Switching. Data Link Layer: Error Detection and Correction, Framing, Flow and Error Control, HDLC, Point-to-Point Protocol, MAC- Random Access, Controlled Access, Channelization, Ethernet Unit III Network Layer: Functions, Network Layer Services, IP Addresses: IPv4 Header, IP Addresses, Classful and Classless Addressing, Subnetting, Network Address Translation, ARP and RARP, ICMP, IGMP, Unicast and Multicast Routing, IPv6			

Unit IV

Transport layer: Introduction, Services, UDP- Header, Services, Applications, TCP- Services, Header, Connections, Flow and Error Control, Congestion Control, Timers, Options, SCTP

Unit V

Application Layer: DHCP, DNS, FTP and Telnet, WWW, HTTP, Electronic Mail

Security: Fundamentals of Network Security, Encryption, Cryptography, SSL, IPsec, Firewalls and Intrusion Detection Systems.

SUGGESTED READINGS

1. Andrew S. Tanenbaum, "Computer Networks", 5th Edition, Pearson Education
2. William Stallings, "Data and Computer Communication", 10th Edition, Pearson Education
3. James F. Kurose & Keith W. Ross, "Computer Networking: A Top –down approach", 6th Edition, Pearson Education
4. Behrouz A. Forouzan, "Data Communications and Networking", 5th Edition, McGraw-Hill Education

C. CO-PO & CO-PSO MAPPING TABLE

COP O	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
CO2	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0
CO3	2	2	1	1	1	0	1	0	0	0	0	1	1	1	0
CO4	2	2	1	1	1	0	1	0	0	0	0	1	1	1	0
CO5	2	1	0	0	1	0	0	1	0	0	0	1	1	0	0

D. THEORY LECTURE PLAN

S.N o.	CONTENT	NUMBER OF LECTURES	Unit
1	Computer Networks, Data Flow	1	Unit-I (04)
2	Network Topology	1	
3	OSI Reference Model	1	
4	TCP/IP Reference Model	1	
5	Analog and Digital Data, Data Rate Limits	1	
6	Performance Analysis, Line Coding		

		1	Unit-II (10)
7	Bandwidth Utilizations	1	
8	Transmission Media, Switching.	1	
CLASS TEST-I			
9	Error Detection and Correction, Framing	2	Unit-III (10)
10	Flow and Error Control, HDLC, Point-to-Point Protocol	2	
11	MAC- Random Access, Controlled Access, Channelization, Ethernet	2	
	MID SEMESTER		
12	Functions, Network Layer Services,	1	
13	IPv4 Header	1	
14	IP Addresses, Classful and Classless Addressing, Subnetting,	2	
15	Network Address Translation, ARP and RARP	2	Unit-IV (10)
16	ICMP	1	
17	IGMP	1	
18	Unicast and Multicast Routing , IPv6	2	
19	Introduction, Services,	1	
20	UDP- Header, Services, Applications,	1	
22	TCP- Services, Header, Connections,	2	
23	Flow and Error Control,	1	Unit-V (06)
24	Congestion Control,	1	
25	Timers, Options,	2	
26	SCTP	2	
27	DHCP, DNS,	1	
28	FTP and Telnet,	1	
29	WWW, HTTP, Electronic Mail	2	
30	Fundamentals of Network Security, Encryption, Cryptography, SSL, IPsec,	1	
31	Firewalls and Intrusion Detection Systems.	1	

E. LAB PLAN

Lab Class No.	Name of the Experiment
1	Understanding Network Commands Execute 20 basic network commands

2 - 6	Packet Tracer ● Learn how to create a simple Local Area Network (LAN) with basic devices like PCs, switches, and routers. ● Understand how to configure a router with basic settings to enable inter-network communication. ● Learn how to set up a DHCP server in Packet Tracer to automatically assign IP addresses to network devices. ● Learn how to manually configure static routes on routers to control network traffic paths. ● Understand how NAT works to enable private IP addresses to communicate with external networks.	
MID SEMESTER EVALUATION		
7-10	Wireshark ● Familiarize with the Wireshark interface, basic functionalities, and how to capture live network traffic. ● Understand how the Address Resolution Protocol (ARP) works in a network to map IP addresses to MAC addresses. ● Learn about the Internet Control Message Protocol (ICMP) and how ping requests are used for network diagnostics. ● Understand the DHCP process for dynamically assigning IP addresses to devices in a network. ● Analyzing the Functionality of Switches through MAC Address Learning ● Examine the effects of network congestion and how packet loss occurs in a network.	
11-12	Mini Project	Project Work & Case Study Work on a mini project using Computer Networks techniques Possible topics: ● Network Traffic Analysis & Packet Sniffing ● Virtual LAN (VLAN) and Network Segmentation ● Dynamic Routing Simulation ● Secure Web Server with TLS
END SEMESTER EVALUATION		

F. SELF STUDY

Sr. No.	Topic	
1	Use Wireshark to analyze movement of data packets	Unit II
2	Study of Packet Tracer for simulations of various computer network algorithms.	Unit III